ALUMINUM

5052

MATERIAL / REFERENCE CONDITION

O · H32 · H34

Typical applications Formed covers, chassis and sheet-metal brackets.

Reference product form	Sheet & plate
Data notes	H32 sheet/plate: tensile 215-265 MPa; 0.2% proof strength >= 160 MPa. Elongation depends on thickness. ASTM B209M reference in source.

Original technical reference | Atlas Steels · external reference

View 5052 data sheet (PDF)

atlassteels.com.au | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

ALUMINUM

6061

MATERIAL / REFERENCE CONDITION

T6 · T651

Typical applications Machined housings, fixtures and mounting plates.

Reference product form	Sheet & plate
Data notes	T6/T651 typical values: tensile 310 MPa; yield 276 MPa; Brinell hardness 95. Typical data, not guaranteed acceptance limits.

Original technical reference | Kaiser Aluminum · external reference

View 6061 data sheet (PDF)

online.kaiseraluminum.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

ALUMINUM

7075

MATERIAL / REFERENCE CONDITION

T6 · T651 · T7351

Typical applications Precision parts requiring a high-strength aluminum grade.

Reference product form	Sheet & plate
Data notes	Select the temper before comparing strength and corrosion performance. Use the source table for the specified product condition.

Original technical reference | Kaiser Aluminum · external reference

View 7075 data sheet (PDF)

online.kaiseraluminum.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

STAINLESS STEEL

304

MATERIAL / REFERENCE CONDITION

Specify delivery condition

Typical applications General-purpose corrosion-resistant covers, brackets and fabricated parts.

Reference product form	Coil · sheet · plate
Data notes	Core 304 / 1.4301 cold-rolled reference: tensile 540-750 MPa; 0.2% proof strength >= 230 MPa at 20°C, EN 10088-2. Do not apply these sheet values to bar stock.

Original technical reference | Outokumpu · external reference

View 304 data sheet (PDF)

www.outokumpu.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

STAINLESS STEEL

316

MATERIAL / REFERENCE CONDITION

Specify delivery condition

Typical applications Fluid-handling and corrosion-resistant components.

Reference product form	Coil · sheet · plate
Data notes	Molybdenum-bearing stainless steel. Match the source product form and specification to the drawing; bar-stock properties require a separate mill datasheet.

Original technical reference | Outokumpu · external reference

View 316 data sheet (PDF)

www.outokumpu.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

STAINLESS STEEL

304L

MATERIAL / REFERENCE CONDITION

Specify delivery condition

Typical applications Welded fabrications requiring a low-carbon grade.

Reference product form	Coil · sheet · plate
Data notes	Low-carbon grade; confirm whether the drawing requires 304L or dual certification. Listed designations are not permission to substitute.

Original technical reference | Outokumpu · external reference

View 304L data sheet (PDF)

www.outokumpu.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

STAINLESS STEEL

316L

MATERIAL / REFERENCE CONDITION

Specify delivery condition

Typical applications Welded fluid-handling parts and corrosion-resistant assemblies.

Reference product form	Coil · sheet · plate
Data notes	Low-carbon molybdenum-bearing grade. Confirm service environment, stock condition and required documentation.

Original technical reference | Outokumpu · external reference

View 316L data sheet (PDF)

www.outokumpu.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

ALUMINUM

6063

MATERIAL / REFERENCE CONDITION

T5 · T6

Typical applications Heat sinks, frames and profiles with visible anodized surfaces.

Reference product form	Extruded profiles
Data notes	Extrusion properties depend on wall thickness and temper. Use the relevant T5 or T6 row in the source; do not substitute sheet or plate values.

Original technical reference | Hydro · external reference

View 6063 data sheet (PDF)

www.hydro.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

ALUMINUM

3003

MATERIAL / REFERENCE CONDITION

Specify temper

Typical applications Formed sheet components and covers.

Reference product form	Sheet & coil
Data notes	Specify sheet thickness, temper and bend requirements. Consult the source for composition and condition-specific properties.

Original technical reference | Atlas Steels · external reference

View 3003 data sheet (PDF)

atlassteels.com.au | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

ALUMINUM

5083

MATERIAL / REFERENCE CONDITION

Specify temper

Typical applications Fabricated aluminum parts and welded structures.

Reference product form	Sheet & plate
Data notes	Confirm temper, thickness and service environment before selecting this grade. Consult the source for condition-specific values.

Original technical reference | Atlas Steels · external reference

View 5083 data sheet (PDF)

atlassteels.com.au | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

CARBON & ALLOY STEEL

Q235B

MATERIAL / REFERENCE CONDITION

Carbon structural steel

Typical applications Welded frames, base plates and brackets

Material type	Carbon structural steel
Condition / reference grade	Hot rolled; specify thickness
Manufacturing	Cutting · welding · machining
Reference data	Reference: GB/T 700; mechanical limits depend on thickness.
Specification notes	Specify welding requirements, impact testing and protective coating.

Original technical reference | Shougang

Mill handbook (PDF)

www.shougang.com.cn | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

CARBON & ALLOY STEEL

Q355B

MATERIAL / REFERENCE CONDITION

Low-alloy structural steel

Typical applications Load-bearing frames and fabricated supports

Material type	Low-alloy structural steel
Condition / reference grade	Hot rolled; specify thickness
Manufacturing	Cutting · welding · machining
Reference data	Shougang hot-rolled reference, thickness <= 16 mm: yield >= 355 MPa; tensile 470-630 MPa.
Specification notes	For thicker sections use the applicable source row; do not treat Q345B as an automatic substitute.

Original technical reference | Shougang

Mill handbook (PDF)

www.shougang.com.cn | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

CARBON & ALLOY STEEL

20 steel

MATERIAL / REFERENCE CONDITION

Low-carbon quality steel

Typical applications Pins, sleeves and lightly loaded machine parts

Material type	Low-carbon quality steel
Condition / reference grade	Specify normalized, annealed or case-hardened condition
Manufacturing	CNC turning · milling · grinding
Reference data	Grade directory reference; request a mill sheet for the specified stock form and treatment. No single strength value applies to every condition.
Specification notes	Specify case depth and core properties if carburizing is required.

Original technical reference | Baosteel

Mill grade directory

esales.baosteel.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

CARBON & ALLOY STEEL

45 steel

MATERIAL / REFERENCE CONDITION

Medium-carbon quality steel

Typical applications Shafts, gears and machine fixtures

Material type	Medium-carbon quality steel
Condition / reference grade	Specify normalized or quenched-and-tempered condition
Manufacturing	CNC turning · milling · grinding
Reference data	Grade directory reference; request a mill sheet for the specified stock form and treatment. No single strength value applies to every condition.
Specification notes	State final hardness, section size and whether induction hardening is required.

Original technical reference | Baosteel

Mill grade directory

esales.baosteel.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

CARBON & ALLOY STEEL

40Cr

MATERIAL / REFERENCE CONDITION

Chromium alloy steel

Typical applications Transmission shafts, gears and couplings

Material type	Chromium alloy steel
Condition / reference grade	Specify quenching and tempering; state final hardness
Manufacturing	CNC turning · milling · grinding
Reference data	Grade directory reference; request a mill sheet for the specified stock form and treatment. No single strength value applies to every condition.
Specification notes	Mechanical properties must match section size and heat treatment.

Original technical reference | Baosteel

Mill grade directory

esales.baosteel.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

CARBON & ALLOY STEEL

42CrMo

MATERIAL / REFERENCE CONDITION

Chromium-molybdenum alloy steel

Typical applications Loaded shafts, spindles and precision machine parts

Material type	Chromium-molybdenum alloy steel
Condition / reference grade	Specify quenching and tempering; state final hardness
Manufacturing	CNC turning · milling · grinding
Reference data	Grade directory reference; request a mill sheet for the specified stock form and treatment. No single strength value applies to every condition.
Specification notes	Do not substitute 4140 or 42CrMo4 without checking the drawing and composition limits.

Original technical reference | Baosteel

Mill grade directory

esales.baosteel.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

SHEET STEEL

SPCC

MATERIAL / REFERENCE CONDITION

Cold-rolled steel sheet

Typical applications Amplifier chassis, equipment covers and sheet-metal brackets

Material type	Cold-rolled steel sheet
Condition / reference grade	Specify temper and surface finish
Manufacturing	Laser cutting · bending · punching
Reference data	Specify sheet thickness in mm, coating designation and required mechanical properties.
Specification notes	Specify SPCCT if a tensile guarantee is required; sheet thickness and temper must be agreed.

Original technical reference | Yodoko

Mill product reference

www.yodoko.co.jp | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

SHEET STEEL

SECC

MATERIAL / REFERENCE CONDITION

Electrogalvanized steel sheet

Typical applications Amplifier chassis, equipment covers and sheet-metal brackets

Material type	Electrogalvanized steel sheet
Condition / reference grade	Specify zinc coating, passivation and oiling
Manufacturing	Laser cutting · bending · punching
Reference data	Specify sheet thickness in mm, coating designation and required mechanical properties.
Specification notes	SECC is not a deep-drawing grade. Coating mass and cut-edge protection are separate requirements.

Original technical reference | Nippon Steel

Mill product reference

www.nipponsteel.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

SHEET STEEL

SGCC

MATERIAL / REFERENCE CONDITION

Hot-dip galvanized steel sheet

Typical applications Amplifier chassis, equipment covers and sheet-metal brackets

Material type	Hot-dip galvanized steel sheet
Condition / reference grade	Specify coating mass and surface appearance
Manufacturing	Laser cutting · bending · punching
Reference data	Specify sheet thickness in mm, coating designation and required mechanical properties.
Specification notes	Confirm bendability and the required coating designation; protect cut edges where needed.

Original technical reference | China Steel & Nippon Steel Vietnam

Mill product reference

www.csvc.com.vn | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

STAINLESS STEEL

303

MATERIAL / REFERENCE CONDITION

Free-machining austenitic stainless

Typical applications Turned fittings and threaded components

Material type	Free-machining austenitic stainless
Condition / reference grade	Specify bar condition
Manufacturing	CNC machining · grinding
Reference data	Full composition, mechanical and thermal property tables in the named grade section of the Atlas reference.
Specification notes	Choose for machining; review welding and corrosion requirements separately.

Original technical reference | Atlas Steels

Grade data sheets (PDF)

www.atlassteels.com.au | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

STAINLESS STEEL

430

MATERIAL / REFERENCE CONDITION

Ferritic stainless

Typical applications Decorative covers and equipment panels

Material type	Ferritic stainless
Condition / reference grade	Annealed sheet; specify finish
Manufacturing	CNC machining · grinding
Reference data	Full composition, mechanical and thermal property tables in the named grade section of the Atlas reference.
Specification notes	Magnetic grade; review corrosion exposure and forming requirements.

Original technical reference | Atlas Steels

Grade data sheets (PDF)

www.atlassteels.com.au | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

STAINLESS STEEL

420

MATERIAL / REFERENCE CONDITION

Martensitic stainless

Typical applications Wear parts, blades and hardened components

Material type	Martensitic stainless
Condition / reference grade	Specify annealed stock and final heat treatment
Manufacturing	CNC machining · grinding
Reference data	Full composition, mechanical and thermal property tables in the named grade section of the Atlas reference.
Specification notes	Carbon level and final hardness must be specified; 420 covers multiple compositions.

Original technical reference | Atlas Steels

Grade data sheets (PDF)

www.atlassteels.com.au | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

STAINLESS STEEL

440C

MATERIAL / REFERENCE CONDITION

High-carbon martensitic stainless

Typical applications Bearing and wear components

Material type	High-carbon martensitic stainless
Condition / reference grade	Specify annealed or hardened condition
Manufacturing	CNC machining · grinding
Reference data	Full composition, mechanical and thermal property tables in the named grade section of the Atlas reference.
Specification notes	Grinding allowance and final hardness depend on heat treatment.

Original technical reference | Atlas Steels

Grade data sheets (PDF)

www.atlassteels.com.au | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

STAINLESS STEEL

17-4PH

MATERIAL / REFERENCE CONDITION

Precipitation-hardening stainless

Typical applications High-strength shafts and valve hardware

Material type	Precipitation-hardening stainless
Condition / reference grade	Specify Condition A, H900, H1025 or H1150
Manufacturing	CNC machining · grinding
Reference data	Full composition, mechanical and thermal property tables in the named grade section of the Atlas reference.
Specification notes	Aging condition changes strength and toughness; do not use one condition's data for another.

Original technical reference | Atlas Steels

Grade data sheets (PDF)

www.atlassteels.com.au | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

COPPER & BRASS

H59

MATERIAL / REFERENCE CONDITION

Brass

Typical applications Machined fittings and general brass hardware

Material type	Brass
Condition / reference grade	Specify extruded or drawn condition
Manufacturing	Turning · milling
Reference data	Cu 57.0-60.0%; Pb <= 0.5%; Fe <= 0.3%; Zn balance. Selected composition only; see source for all limits.
Specification notes	Do not describe this grade as lead-free.

Original technical reference | Jintian Copper

Mill composition table

jtcopper.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

COPPER & BRASS

H62

MATERIAL / REFERENCE CONDITION

Brass

Typical applications Formed brass components and decorative parts

Material type	Brass
Condition / reference grade	Specify extruded or drawn condition
Manufacturing	Turning · milling
Reference data	Cu 60.5-63.5%; Pb <= 0.08%; Fe <= 0.15%; Zn balance. Selected composition only; see source for all limits.
Specification notes	Specify temper; strength and formability change with cold work.

Original technical reference | Jintian Copper

Mill composition table

jtcopper.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

COPPER & BRASS

HPb59-1

MATERIAL / REFERENCE CONDITION

Brass

Typical applications Turned inserts, connectors and threaded fittings

Material type	Brass
Condition / reference grade	Specify extruded or drawn condition
Manufacturing	Turning · milling
Reference data	Cu 57.0-60.0%; Pb 0.8-1.9%; Fe <= 0.5%; Zn balance. Selected composition only; see source for all limits.
Specification notes	Leaded brass: identify any lead-content restrictions in the RFQ.

Original technical reference | Jintian Copper

Mill composition table

jtcopper.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

COPPER & BRASS

T2

MATERIAL / REFERENCE CONDITION

Copper

Typical applications Electrical contacts and heat-transfer components

Material type	Copper
Condition / reference grade	Specify annealed, half-hard or hard condition
Manufacturing	Machining · forming
Reference data	Jintian publishes grade and temper-specific data. Strip data must not be used as bar-stock acceptance limits.
Specification notes	Specify conductivity, temper and joining requirements.

Original technical reference | Jintian Copper

Mill strip catalogue (PDF)

jtcopper.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

COPPER & BRASS

TU1

MATERIAL / REFERENCE CONDITION

Oxygen-free copper

Typical applications Electrical and thermal components requiring low oxygen content

Material type	Oxygen-free copper
Condition / reference grade	Specify annealed, half-hard or hard condition
Manufacturing	Machining · forming
Reference data	Jintian publishes grade and temper-specific data. Strip data must not be used as bar-stock acceptance limits.
Specification notes	Confirm oxygen and purity limits from the actual mill certificate.

Original technical reference | Jintian Copper

Mill strip catalogue (PDF)

jtcopper.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

TOOL STEEL

P20

MATERIAL / REFERENCE CONDITION

Plastic mould steel

Typical applications Injection moulds and mould inserts

Material type	Plastic mould steel
Condition / reference grade	Specify prehardened condition and mill grade
Manufacturing	Milling · EDM · grinding
Reference data	Use the supplier's P20 datasheet; the linked Impax reference is a specific Ni-modified mould steel.
Specification notes	The branded reference is not a guarantee for generic P20.

Original technical reference | Uddeholm

Technical reference

www.uddeholm.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

TOOL STEEL

718 mould steel

MATERIAL / REFERENCE CONDITION

Ni-modified plastic mould steel

Typical applications Larger mould blocks and polished cavities

Material type	Ni-modified plastic mould steel
Condition / reference grade	Specify manufacturer and hardness
Manufacturing	Milling · EDM · grinding
Reference data	Reference example: Impax Supreme / AS 718 Supreme supplied at 290-330 HB.
Specification notes	This is mould steel, not Inconel 718 nickel alloy.

Original technical reference | Uddeholm

Technical reference

www.uddeholm.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

TOOL STEEL

H13

MATERIAL / REFERENCE CONDITION

Hot-work tool steel

Typical applications Die-casting tooling and hot-work inserts

Material type	Hot-work tool steel
Condition / reference grade	Annealed stock; final hardened and tempered condition to drawing
Manufacturing	Milling · EDM · grinding
Reference data	Reference example: Orvar Supreme, AISI H13 / 1.2344.
Specification notes	Specify final hardness and heat-treatment quality; branded ESR data is not generic H13 certification.

Original technical reference | Uddeholm

Technical reference

www.uddeholm.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

TOOL STEEL

Cr12MoV

MATERIAL / REFERENCE CONDITION

Cold-work tool steel

Typical applications Blanking dies and wear inserts

Material type	Cold-work tool steel
Condition / reference grade	Annealed stock; final hardened and tempered condition to drawing
Manufacturing	Milling · EDM · grinding
Reference data	Selected composition: C 1.45-1.70%; Cr 11.00-12.50%; Mo 0.40-0.60%; V 0.15-0.30%.
Specification notes	Confirm the full specification; do not automatically substitute D2 or SKD11.

Original technical reference | Fushun Metal

Technical reference

www.fushunspecialsteel.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

CASTING ALLOYS

ADC12

MATERIAL / REFERENCE CONDITION

Aluminum die-casting alloy

Typical applications Thin-wall housings and intricate cast components

Material type	Aluminum die-casting alloy
Condition / reference grade	As-cast; specify any secondary treatment
Manufacturing	Die casting · secondary machining
Reference data	Use the ADC12 specification; do not replace it with A380 based only on similar applications.
Specification notes	Specify porosity, sealing, coating and critical machined surfaces. Reference values are not billet properties.

Original technical reference | Dynacast

Alloy property selector

www.dynacast.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

CASTING ALLOYS

A380

MATERIAL / REFERENCE CONDITION

Aluminum die-casting alloy

Typical applications Equipment housings and cast brackets

Material type	Aluminum die-casting alloy
Condition / reference grade	As-cast; specify any secondary treatment
Manufacturing	Die casting · secondary machining
Reference data	Reference hardness: 80 HB in the Dynacast selector; actual casting properties depend on process and section.
Specification notes	Specify porosity, sealing, coating and critical machined surfaces. Reference values are not billet properties.

Original technical reference | Dynacast

Alloy property selector

www.dynacast.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

CASTING ALLOYS

Zamak 3

MATERIAL / REFERENCE CONDITION

Zinc die-casting alloy

Typical applications Detailed small housings and decorative hardware

Material type	Zinc die-casting alloy
Condition / reference grade	As-cast; specify any secondary treatment
Manufacturing	Die casting · secondary machining
Reference data	Reference hardness: 82 HB in the Dynacast selector; use agreed casting acceptance criteria.
Specification notes	Specify porosity, sealing, coating and critical machined surfaces. Reference values are not billet properties.

Original technical reference | Dynacast

Alloy property selector

www.dynacast.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

CASTING ALLOYS

Zamak 5

MATERIAL / REFERENCE CONDITION

Zinc die-casting alloy

Typical applications Loaded fittings and hardware

Material type	Zinc die-casting alloy
Condition / reference grade	As-cast; specify any secondary treatment
Manufacturing	Die casting · secondary machining
Reference data	Reference hardness: 91 HB in the Dynacast selector; use agreed casting acceptance criteria.
Specification notes	Specify porosity, sealing, coating and critical machined surfaces. Reference values are not billet properties.

Original technical reference | Dynacast

Alloy property selector

www.dynacast.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

ENGINEERING PLASTICS

ABS

MATERIAL / REFERENCE CONDITION

Acrylonitrile butadiene styrene

Typical applications Enclosures and acoustic waveguides

Material type	Acrylonitrile butadiene styrene
Condition / reference grade	POLYLAC PA-757 (example)
Manufacturing	Machining or molding; confirm grade-specific route
Reference data	Choose the PA-757 ISO or ASTM sheet in the manufacturer's product table.
Specification notes	Confirm color, impact requirements and flame rating for the exact grade. Branded examples are references, not a statement of stocked brand.

Original technical reference | CHIMEI

Manufacturer grade reference

www.chimeicorp.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

ENGINEERING PLASTICS

PC

MATERIAL / REFERENCE CONDITION

Polycarbonate

Typical applications Transparent covers and impact-resistant housings

Material type	Polycarbonate
Condition / reference grade	Makrolon 2805 (example)
Manufacturing	Machining or molding; confirm grade-specific route
Reference data	Example MVR: 9 cm3/10 min at 300°C / 1.2 kg, ISO 1133-1.
Specification notes	MVR is a processing property, not a mechanical-strength value. Branded examples are references, not a statement of stocked brand.

Original technical reference | Covestro

Manufacturer grade reference

solutions.covestro.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

ENGINEERING PLASTICS

PC/ABS

MATERIAL / REFERENCE CONDITION

Polycarbonate / ABS blend

Typical applications Electronic enclosures and molded covers

Material type	Polycarbonate / ABS blend
Condition / reference grade	Bayblend T65 XF (example)
Manufacturing	Machining or molding; confirm grade-specific route
Reference data	Example Vicat/B 120: 120°C. This is not a continuous-use temperature.
Specification notes	State grade, color and flame-rating requirements. Branded examples are references, not a statement of stocked brand.

Original technical reference | Covestro

Manufacturer grade reference

solutions.covestro.com | External publisher; reference only

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ENGINEERING PLASTICS

POM

MATERIAL / REFERENCE CONDITION

Acetal copolymer

Typical applications Gears, bushings and sliding parts

Material type	Acetal copolymer
Condition / reference grade	TECAFORM AH natural (example)
Manufacturing	Machining or molding; confirm grade-specific route
Reference data	Example density 1.41 g/cm3; tensile strength 67 MPa, DIN EN ISO 527-2, 50 mm/min.
Specification notes	Stock-shape data; do not transfer directly to another molded resin or POM-H grade. Branded examples are references, not a statement of stocked brand.

Original technical reference | Ensinger

Manufacturer grade reference

www.ensinger-online.com | External publisher; reference only

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ENGINEERING PLASTICS

PA6

MATERIAL / REFERENCE CONDITION

Nylon 6

Typical applications Wear pads, gears and machine parts

Material type	Nylon 6
Condition / reference grade	TECAMID 6 natural (example)
Manufacturing	Machining or molding; confirm grade-specific route
Reference data	Select the exact unfilled or reinforced grade in the technical-sheet directory.
Specification notes	Specify moisture conditioning; dry and conditioned properties differ. Branded examples are references, not a statement of stocked brand.

Original technical reference | Ensinger

Manufacturer grade reference

www.ensinger-online.com | External publisher; reference only

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ENGINEERING PLASTICS

PA66

MATERIAL / REFERENCE CONDITION

Nylon 66

Typical applications Mechanical brackets and wear components

Material type	Nylon 66
Condition / reference grade	TECAMID 66 natural (example)
Manufacturing	Machining or molding; confirm grade-specific route
Reference data	Select the exact unfilled or reinforced grade in the technical-sheet directory.
Specification notes	Specify glass content, moisture conditioning and molding or stock-shape route. Branded examples are references, not a statement of stocked brand.

Original technical reference | Ensinger

Manufacturer grade reference

www.ensinger-online.com | External publisher; reference only

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ENGINEERING PLASTICS

PMMA

MATERIAL / REFERENCE CONDITION

Acrylic

Typical applications Clear covers and optical housings

Material type	Acrylic
Condition / reference grade	ACRYREX CM-205 (example)
Manufacturing	Machining or molding; confirm grade-specific route
Reference data	Manufacturer provides separate ISO and ASTM data sheets for CM-205.
Specification notes	Specify optical requirements, surface finish and stress-cracking exposure. Branded examples are references, not a statement of stocked brand.

Original technical reference | CHIMEI

Manufacturer grade reference

www.chimeicorp.com | External publisher; reference only

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ENGINEERING PLASTICS

PP

MATERIAL / REFERENCE CONDITION

Polypropylene

Typical applications Chemical-handling components and fixtures

Material type	Polypropylene
Condition / reference grade	TECAFINE PP natural (example)
Manufacturing	Machining or molding; confirm grade-specific route
Reference data	Example density: 0.90 g/cm3. The linked reference is a stock-shape grade.
Specification notes	Homopolymer and copolymer grades differ; specify fillers and weld requirements. Branded examples are references, not a statement of stocked brand.

Original technical reference | Ensinger

Manufacturer grade reference

www.ensinger-online.com | External publisher; reference only

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ENGINEERING PLASTICS

PE

MATERIAL / REFERENCE CONDITION

Polyethylene

Typical applications Guides, protective blocks and fabricated parts

Material type	Polyethylene
Condition / reference grade	TECAFINE PE300 natural (example)
Manufacturing	Machining or molding; confirm grade-specific route
Reference data	Example HDPE density: 0.96 g/cm3; tensile modulus 1100 MPa, DIN EN ISO 527-1.
Specification notes	HDPE, LDPE and UHMW-PE are different selections; do not apply this sheet to all PE. Branded examples are references, not a statement of stocked brand.

Original technical reference | Ensinger

Manufacturer grade reference

www.ensinger-online.com | External publisher; reference only

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ENGINEERING PLASTICS

PTFE

MATERIAL / REFERENCE CONDITION

Polytetrafluoroethylene

Typical applications Seals, valve seats and low-friction interfaces

Material type	Polytetrafluoroethylene
Condition / reference grade	TECAFLON PTFE natural (example)
Manufacturing	Machining or molding; confirm grade-specific route
Reference data	Use the TECAFLON PTFE column in the supplier's stock-shape reference.
Specification notes	Specify virgin or filled PTFE; creep and thermal expansion affect fits. Branded examples are references, not a statement of stocked brand.

Original technical reference | Ensinger

Manufacturer grade reference

www.ensingerplastics.com | External publisher; reference only

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ENGINEERING PLASTICS

PEEK

MATERIAL / REFERENCE CONDITION

Polyether ether ketone

Typical applications High-performance bushings and insulating parts

Material type	Polyether ether ketone
Condition / reference grade	TECAPEEK natural (example)
Manufacturing	Machining or molding; confirm grade-specific route
Reference data	Example tensile strength: 116 MPa; tensile modulus: 4200 MPa (ISO 527).
Specification notes	Unfilled reference; glass/carbon-filled grades need their own sheet. Branded examples are references, not a statement of stocked brand.

Original technical reference | Ensinger

Manufacturer grade reference

www.ensingerplastics.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

ELASTOMERS

Silicone

MATERIAL / REFERENCE CONDITION

Silicone elastomer / VMQ

Typical applications Molded seals and flexible interfaces

Material type	Silicone elastomer / VMQ
Condition / reference grade	ELASTOSIL LR 3003/50 (example)
Manufacturing	Molding · application-specific finishing
Reference data	Example: 50 Shore A, density 1.13 g/cm3, tensile strength 10.3 MPa (ISO 37 Type 1).
Specification notes	Values apply to the named cured compound. Specify curing and post-curing requirements. Confirm the selected compound before production.

Original technical reference | WACKER

Compound reference

www.wackerconnect.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

ELASTOMERS

EPDM

MATERIAL / REFERENCE CONDITION

Ethylene propylene diene rubber

Typical applications Weather-exposed and water-service seals

Material type	Ethylene propylene diene rubber
Condition / reference grade	Specify compound code and Shore A hardness
Manufacturing	Molding · application-specific finishing
Reference data	Material-family guide: select the compound and consult its test data.
Specification notes	Confirm fluid compatibility; EPDM is generally unsuitable for mineral oils. Confirm the selected compound before production.

Original technical reference | Parker

Compound reference

www.parker.com | External publisher; reference only

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ELASTOMERS

NBR

MATERIAL / REFERENCE CONDITION

Nitrile rubber

Typical applications Oil-service seals and gaskets

Material type	Nitrile rubber
Condition / reference grade	Specify compound code and Shore A hardness
Manufacturing	Molding · application-specific finishing
Reference data	Material-family guide: acrylonitrile content and compound formulation influence performance.
Specification notes	Review oil, temperature, ozone exposure and compression set. Confirm the selected compound before production.

Original technical reference | Parker

Compound reference

www.parker.com | External publisher; reference only

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ELASTOMERS

FKM

MATERIAL / REFERENCE CONDITION

Fluoroelastomer

Typical applications Heat- and chemical-exposed seals

Material type	Fluoroelastomer
Condition / reference grade	Specify compound code, cure system and hardness
Manufacturing	Molding · application-specific finishing
Reference data	Material-family guide: polymer type and cure system influence fluid compatibility.
Specification notes	Specify actual chemicals and temperature; FKM is not universally compatible. Confirm the selected compound before production.

Original technical reference | Parker

Compound reference

www.parker.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.

ELASTOMERS

TPU

MATERIAL / REFERENCE CONDITION

Thermoplastic polyurethane

Typical applications Flexible wear parts, protective covers and seals

Material type	Thermoplastic polyurethane
Condition / reference grade	Specify exact Elastollan grade (example family)
Manufacturing	Molding · application-specific finishing
Reference data	BASF range tables separate hardness, density, tensile and elongation by product code.
Specification notes	Choose polyester or polyether chemistry and Shore A/D hardness; do not combine values from different variants. Confirm the selected compound before production.

Original technical reference | BASF

Compound reference

download.basf.com | External publisher; reference only

Selection reference, not a material certificate or guaranteed purchasing specification. Properties apply only to the stated grade, form, condition and test method. Branded examples do not establish Metalcorex stock or supplier identity. Confirm the production specification, supplier sheet and required records with your quotation.